

LLM as a Clustering Oracle: Necessary at Every Stage?

Testing the robustness of LLM-guided text clustering before, during, and after partitioning

Article: Large Language Models Enable Few-Shot Clustering

Viswanathan et al. (2024) — *TACL*

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Supervisor: Lazhar Labiod

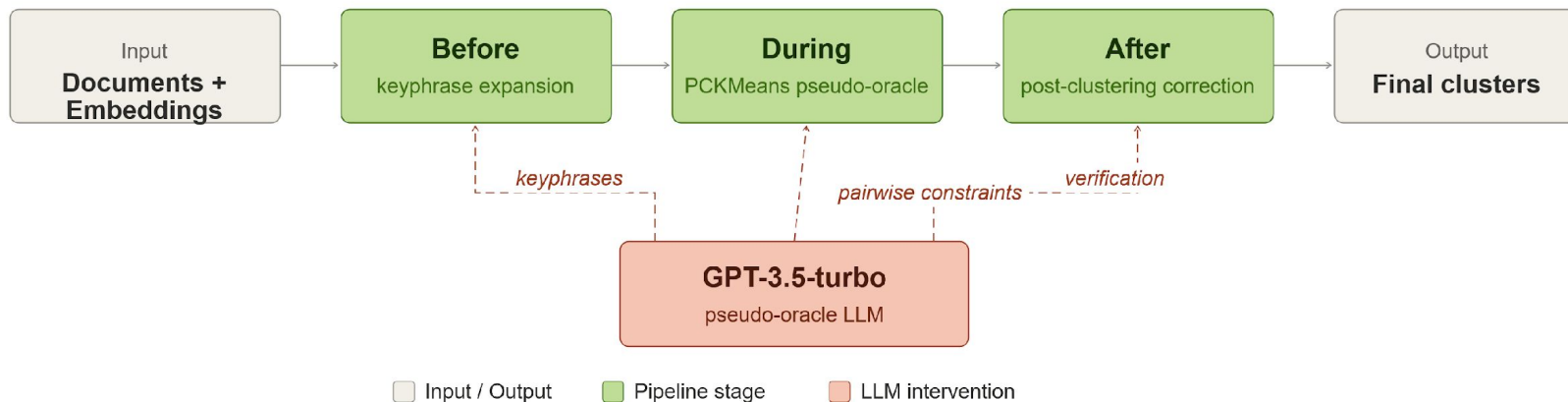


QR → paper

THE KEY IDEA

A three-stage framework: inject LLM signal before, during and after clustering

Clustering is under-specified, the LLM replaces the human oracle to disambiguate user intent



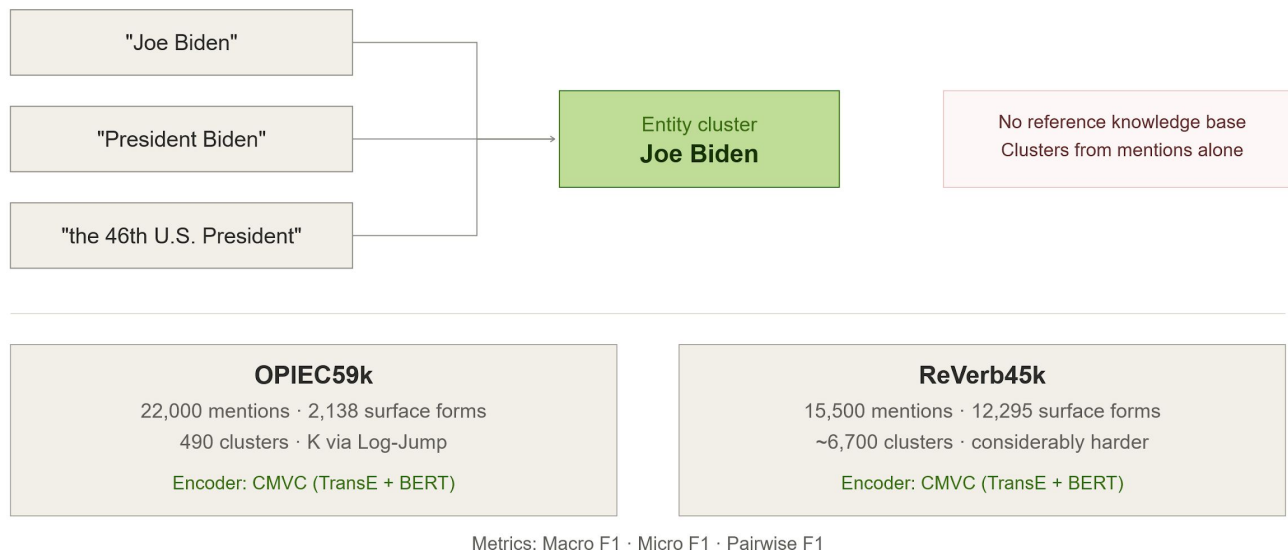
*“The LLM as a pseudo-oracle for text clustering: genuine semantic contribution,
or substitutable technical compensation?”*

Four evaluations, one per stage

Stage	Article's claim	Our evaluation	Our question
Before Keyphrase expansion	The LLM enriches document representations	Eval 1 Modern encoder	Gain from LLM or from a dated encoder?
During PCKMeans oracle	The LLM produces reliable pairwise constraints	Eval 2 Transitive inconsistencies	Are the 20k constraints logically coherent?
After Post-clustering fix	The LLM corrects residual misassignments	Eval 3 Geometric baseline	LLM needed or does geometry suffice?
Cross-cutting Generality	The approach is encoder-agnostic and general	Eval 4 20 Newsgroups dataset	Conclusions hold out-of-distribution?

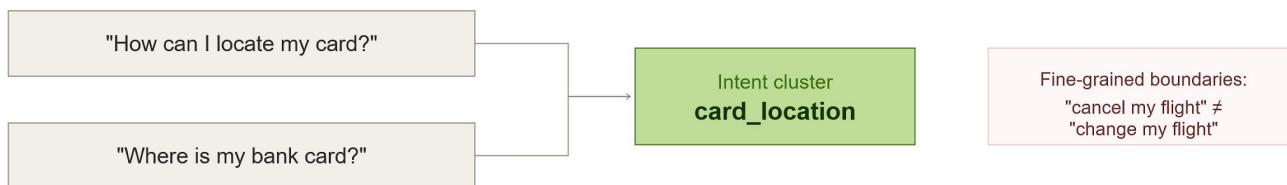
TASK 1 — ENTITY CANONICALIZATION

Group mentions that refer to the same real-world entity



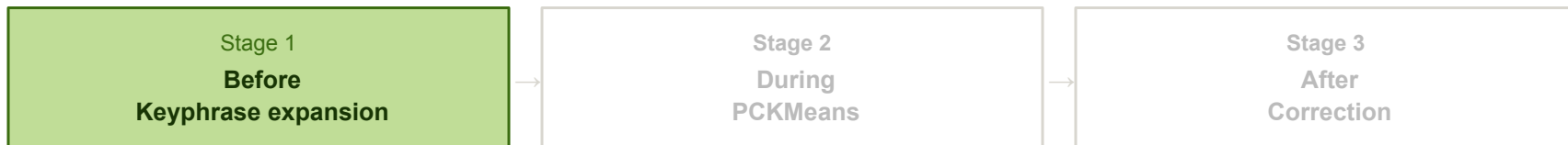
TASK 2 — INTENT CLUSTERING

Group short utterances by the intention they express



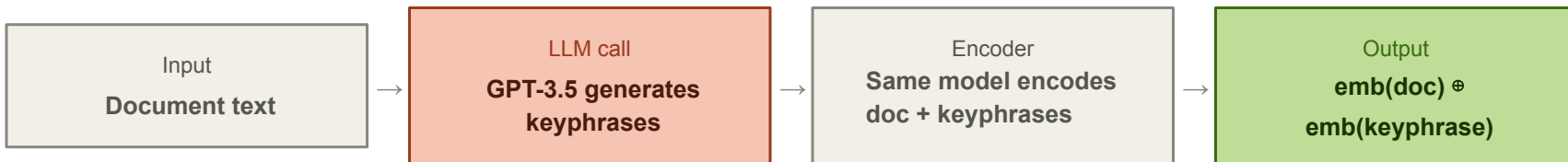
Metrics: Accuracy (Hungarian alignment) · Normalized Mutual Information

STAGE 1 · ENRICH



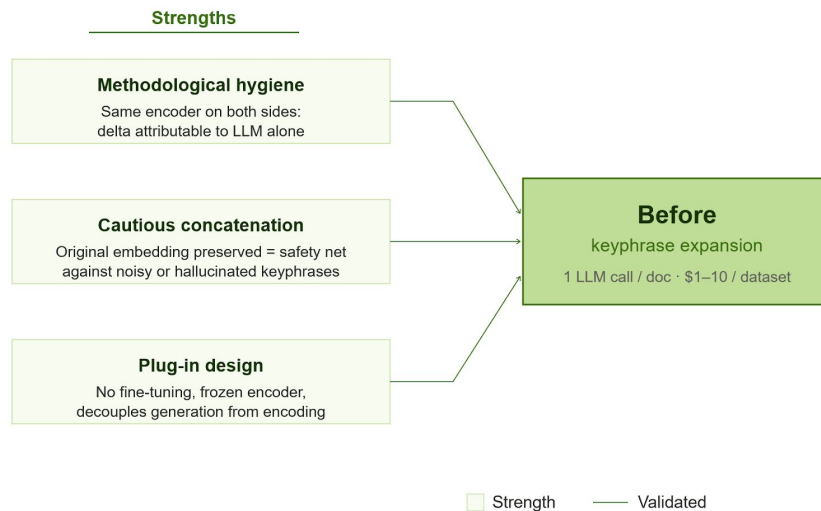
**Does the gain from keyphrase expansion come
from the LLM or from a dated encoder?**

The LLM generates keyphrases to enrich frozen document embeddings



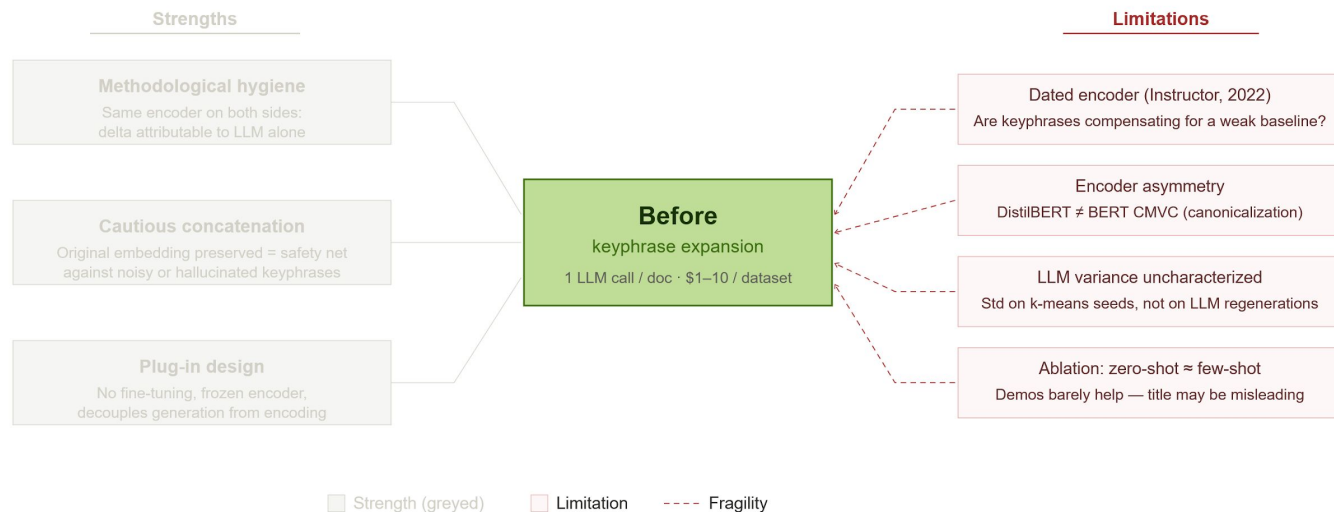
Gains on **3/5 datasets** · CLINC: **+5.8 acc, +3.4 NMI** · Cost: **\$1–10 per dataset** · Best cost-performance ratio of the three stages

Same encoder, cautious fusion, no fine-tuning

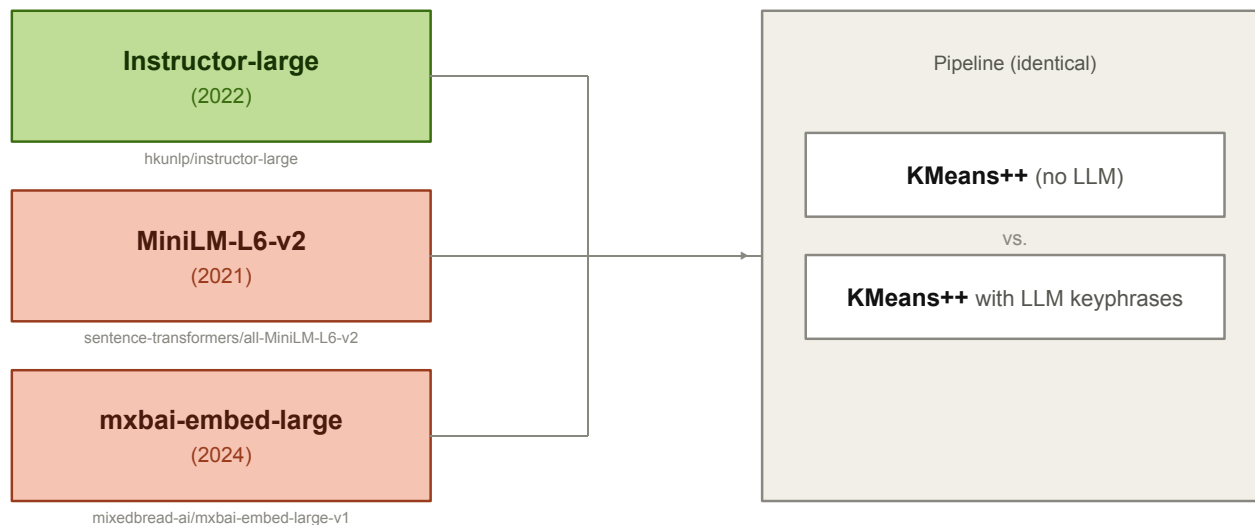


STAGE 1 · ENRICH — LIMITATIONS

Dated encoder, uncharacterized variance, and few-shot in name only



Do keyphrases add semantic value or compensate for weaker encoders?



CLINC · 500 documents · same pipeline, same seeds. Only the encoder changes.

A modern encoder without LLM matches the paper's encoder with LLM

Metric: Accuracy (Hungarian alignment)

Encoder	KMeans (no LLM)	+ LLM keyphrases	Δ
Instructor (2022)	59.0	62.4	+3.4
MiniLM (2021)	62.0	60.6	-1.4
mxbai (2024)	62.2	64.8	+2.6

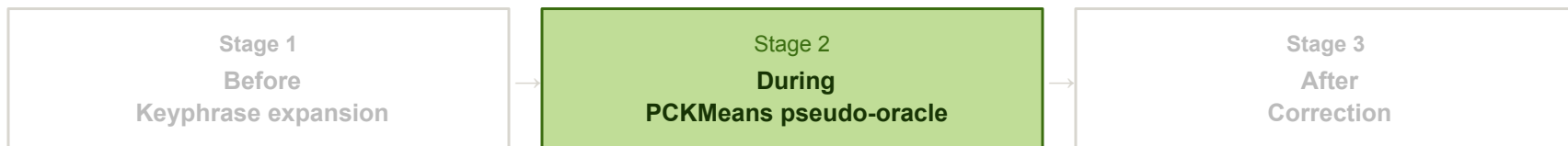
mxbai without LLM (62.2) ~ Instructor with LLM (62.4)

the keyphrase stage compensates for a dated encoder

VERDICT The LLM's value is encoder-dependent, not absolute.

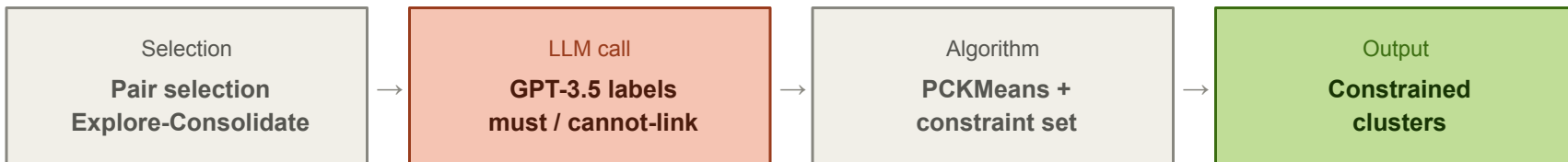
All results: **CLINC, 500-doc subset** · our focused reproduction · **LLM: gpt-5.4-nano**

STAGE 2 · CONSTRAIN



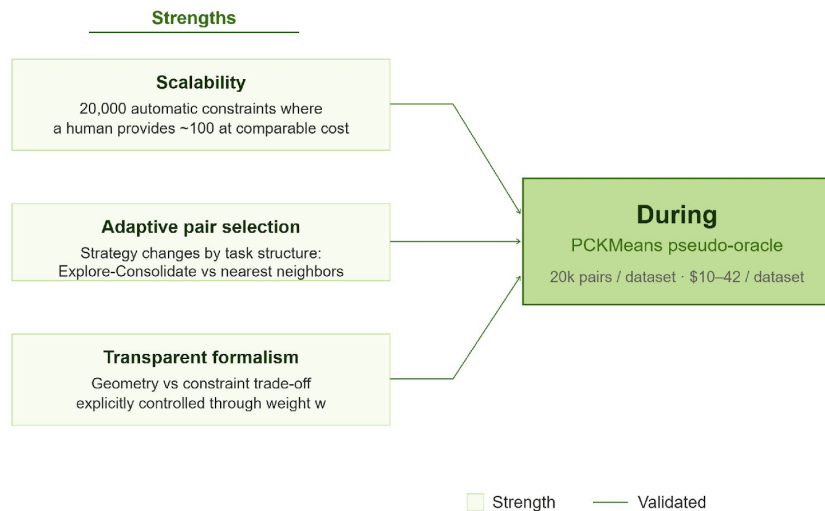
**Are 20,000 LLM pairwise judgments
logically coherent?**

The LLM replaces the human oracle, classifying 20,000 pairwise constraints



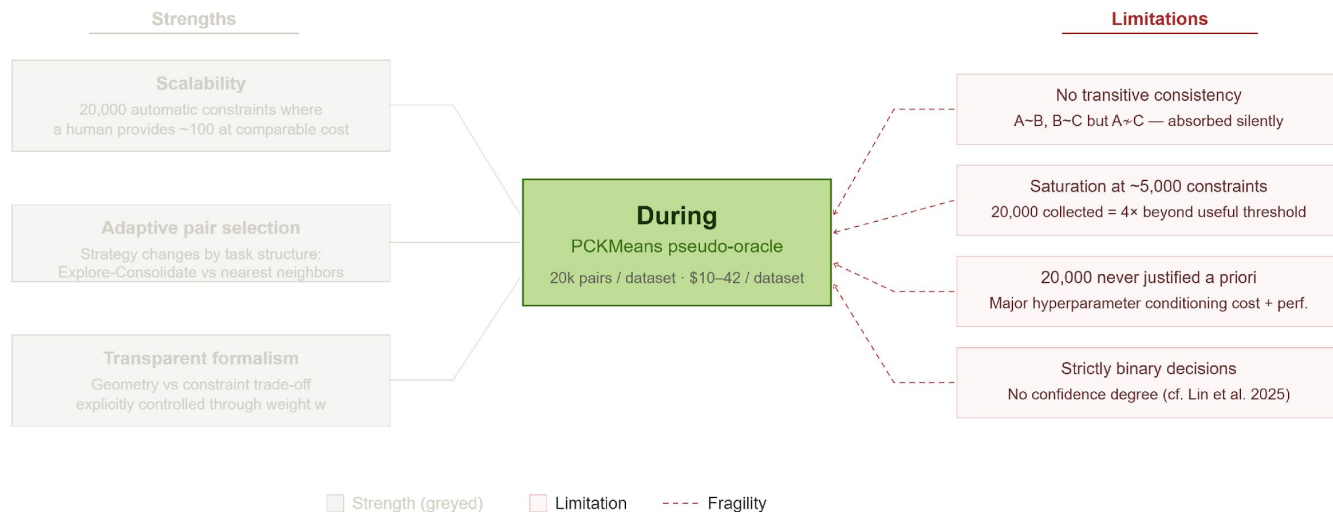
Cost: **\$10–42 per dataset.** ~28× more constraints than a human annotator at equal budget.

Scalability, adaptive selection, and transparent constraint formalism

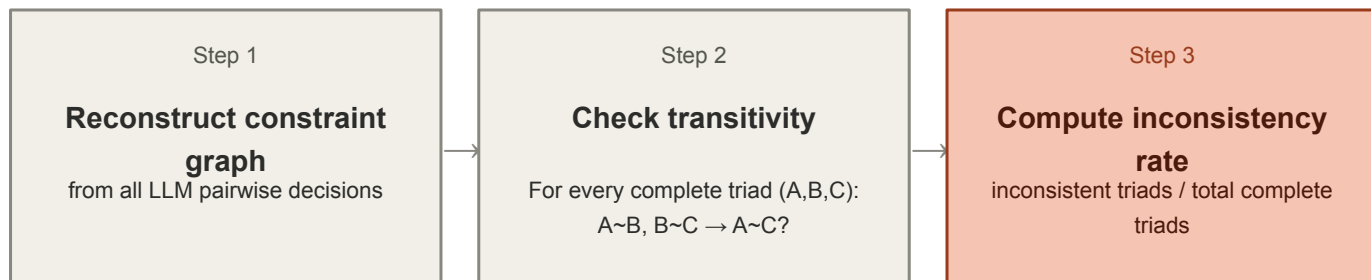


STAGE 2 · CONSTRAIN — LIMITATIONS

No transitive consistency, early saturation, and binary-only decisions



Are the pairwise constraints logically coherent?



CLINC · 500 docs · 1,000 constraints · 2 encoders (Instructor, mxbai) · ~12,800 complete triads each

Zero contradictions but only because the oracle says "no" to everything

0

transitive contradictions across ~25,600 triads

But why?

997 cannot-link · 2 must-link · 1 abstention

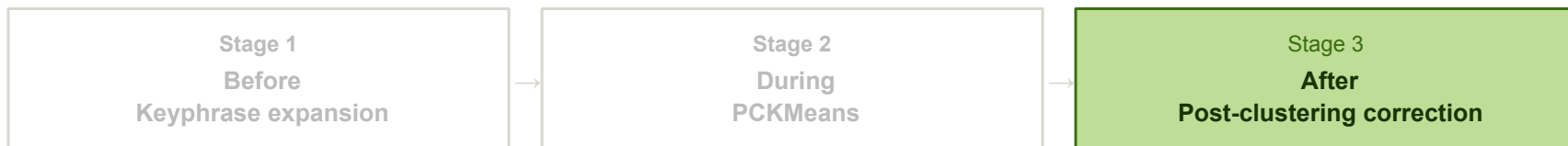
per 1,000 queries · pattern identical on both encoders

A purely cannot-link graph cannot contain a transitive contradiction by construction.

Saturation confirmed: doubling from 500 → 1,000 constraints = zero improvement

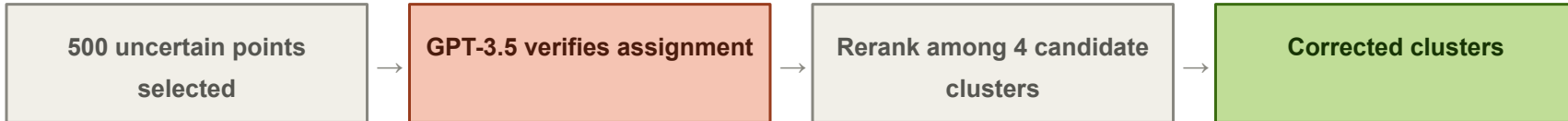
VERDICT The bottleneck is not constraint quantity, but how balanced those judgments are.

STAGE 3 · CORRECT



**Is the LLM actually needed for correction or
does simple geometry suffice?**

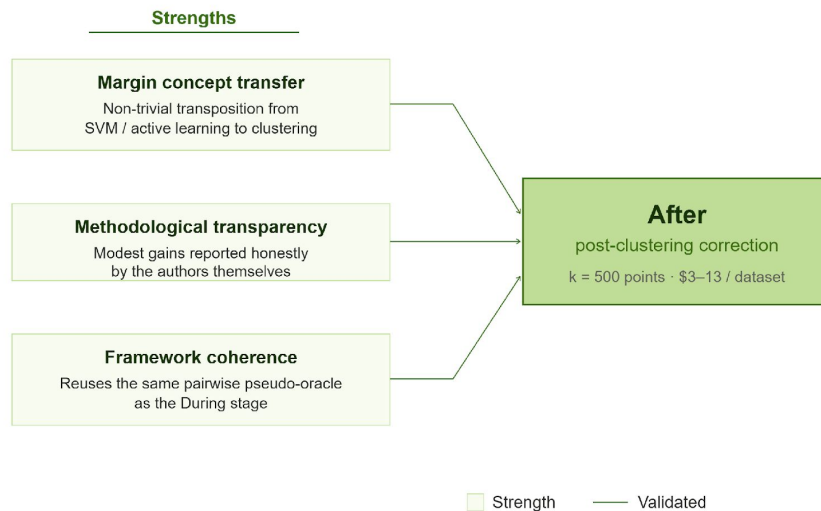
The LLM corrects low-confidence assignments but gains are marginal



Reported gains: **+0.1 to +5.2 points**

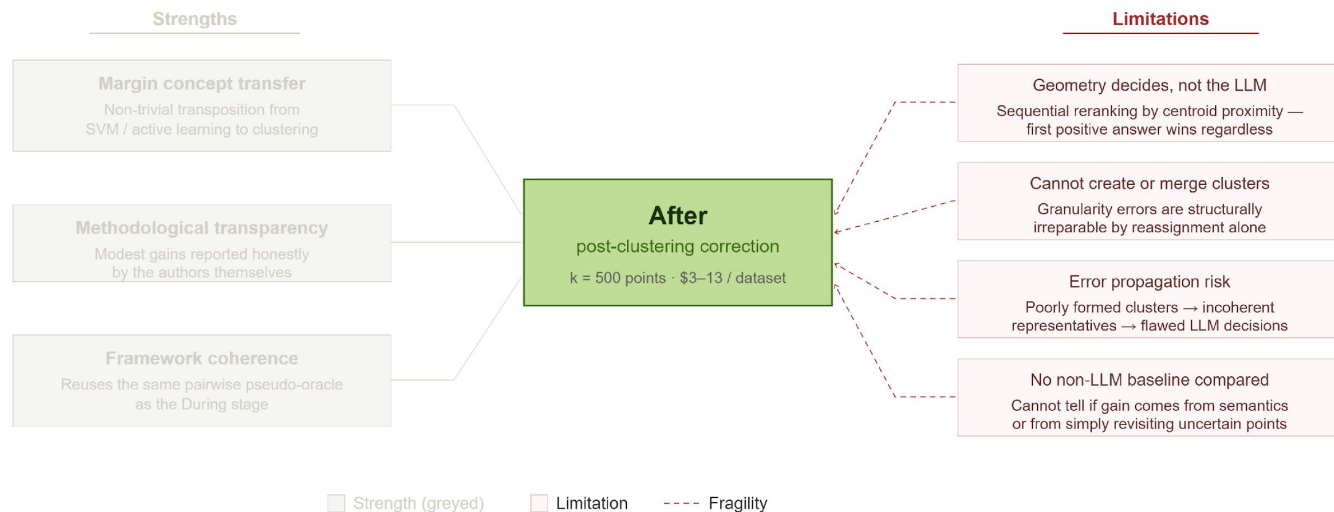
Stage hierarchy: **Before > During > After**

Margin-based targeting, honest reporting, and framework coherence



STAGE 3 · CORRECT — LIMITATIONS

Geometry overrides the LLM, no structural correction, and no non-LLM baseline



Can geometry alone match the LLM correction?



Geometry suffices on the paper's encoder but degrades a strong one

Method	Accuracy	Pairwise F1	Δ vs KMeans
Instructor (2022)			
KMeans	59.0	39.8	—
Geometric correction	62.0	48.0	+3.0 / +8.2
mxbai (2024)			
KMeans	62.2	49.5	—
Geometric correction	55.0	30.8	-7.2 / -18.7

On the paper's encoder, geometry outperforms the LLM at zero cost. On a strong encoder, correction breaks what already works.

VERDICT Post-correction is encoder-dependent regardless of mechanism.

Exact replication is impossible, we chose a focused pilot approach

CONSTRAINTS

Original model gpt-3.5-turbo-0301 no longer accessible

Paper: **full datasets (up to 4,500 docs)**

20,000 constraints · 5 seeds

OUR APPROACH

Substitute: **gpt-5.4-nano** (documented in every run)

500-doc subsets · up to 1,000 constraints

Focus: verify pipeline behavior, concentrate budget on 4 evaluations

Score gaps with the paper are expected and explained by scale differences.

Does the approach generalize beyond the paper's benchmark?

ORIGINAL BENCHMARK

OPIEC59k

ReVerb45k

Banking77

CLINC

Tweet



OUT-OF-DISTRIBUTION

OUR ADDITION

20 Newsgroups

300 docs · 3 topics · TF-IDF · 30 constraints · 3 seeds

Pairwise pseudo-oracle tested under domain shift

The LLM helps out-of-distribution but confidence drops sharply

+6.3

accuracy gain

75.2 → 81.6

40%

oracle abstention rate

pseudo-oracle declined to label

Transferability signal exists, but the pseudo-oracle's reliability degrades outside intent-clustering.

Broader generalization cannot be claimed from a 300-document pilot.

VERDICT Preliminary: the LLM transfers but with degraded confidence.

The gains are real but more contingent than the paper suggests

Enrich

Encoder-dependent. A 2024 encoder without LLM matches Instructor with LLM.

Constrain

Volume without quality. 99.7% cannot-link → trivially consistent but informationally poor.

Correct

Geometry suffices on the paper's encoder. On a strong encoder, both corrections degrade.

Generalize

Signal exists but 40% abstention shows degraded reliability out-of-distribution.

Answer: The three-stage framework remains valuable as an analytical lens but the gains are contingent on encoder, constraint distribution, and initial clustering quality.

Thank you

Questions?

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Viswanathan et al. (2024), TACL doi: 10.1162/tac_l_a_00662 · arXiv: 2307.00524

